

Automated Classification of Atrial Fibrillation from ECG Signals Using Machine Learning Techniques

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Abstract— Atrial fibrillation (AF) is the most prevalent form of cardiac arrhythmia and is strongly associated with an increased risk of stroke and heart failure, which has driven extensive research on its detection using electrocardiogram (ECG) signals. In this study, we propose a machine learning-based automated approach for AF detection from ECG signals. A comprehensive set of time-domain, frequency-domain, and morphological features was extracted from the ECG recordings. Among the three classifiers deployed: Support Vector Machine (SVM), Logistic Regression, and XGBoost—SVM demonstrated the most robust performance, achieving an overall accuracy of 95.00% and 95.60% F1 score. Furthermore, feature importance analysis was conducted to identify the most discriminative attributes contributing to AF detection. The experimental results highlight that the proposed method provides an effective and interpretable solution for automatic AF detection, offering potential clinical utility for early diagnosis and monitoring.

Keywords— Atrial Fibrillation, Electrocardiography, Machine Learning, Signal Processing.

I. INTRODUCTION

Atrial fibrillation (AF) is a persistent and progressive form of cardiac arrhythmia that has attracted considerable global attention due to its rising prevalence and strong association with increased mortality in recent years. Cardiovascular diseases (CVDs) account for nearly 32% of all deaths worldwide, with heart attacks contributing to approximately 85% of these cases [1]. AF significantly increases the risk of thromboembolic events and is linked to nearly 16% of cryptogenic stroke cases [2]. Globally, AF affected about 43.6 million people in 2016, and this number is projected to reach nearly 72 million by 2050 [3]. Early detection of AF with high accuracy is essential to ensure effective clinical management and prevent potentially fatal outcomes. Traditional approaches for AF detection typically rely on clinical inspection of electrocardiogram (ECG) signals, focusing on features such as RR interval irregularities, heart rate variability, and P-wave morphology. While these methods provide useful diagnostic cues, they are often limited by subjectivity and may fail to generalize across large patient populations. To address these challenges, automated diagnostic frameworks have been increasingly investigated, particularly in the context of clinical decision support systems. The availability of publicly accessible ECG databases has further accelerated this trend, allowing researchers to evaluate computational models on diverse signal recordings without the restrictions of small-scale clinical datasets.

Significant research efforts have been devoted to the development of automated AF detection systems for Clinical Decision Support (CDS), facilitated by the growing

accessibility of open-source ECG datasets. Multiple reviews have highlighted the role of deep learning in detecting heart arrhythmias [4]. For example, time-frequency representations such as the short-time Fourier transform have been combined with data augmentation using generative neural networks to improve classifier generalization [5]. Other investigations trained neural networks on long-term ECG recordings and later applied transfer learning on smaller datasets to adapt the models [6]. Similarly, one-dimensional (1-D) signal augmentation strategies have been integrated with convolutional neural networks to enhance performance [7], and novel augmentation approaches for ECG waveforms have also been proposed [8].

In this study, we present a machine learning framework designed to detect AF with improved accuracy and reliability. The methodology involves preprocessing the raw ECG signals to remove noise and baseline wander, followed by feature extraction from time-domain, frequency-domain, and morphology characteristics of the RR intervals and ECG waveform. The extracted features are then standardized and input into multiple supervised classifiers, including Support Vector Machine (SVM), Random Forest (RF), and eXtreme Gradient Boosting (XGBoost), to identify AF episodes. Model performance is evaluated using cross-validation, and metrics such as accuracy, precision, recall, and the area under the Receiver Operating Characteristic (ROC) curve are reported. This approach emphasizes not only high detection accuracy but also the interpretability and robustness of the classifiers for potential clinical application.

II. MATERIAL AND METHODS

A. Database

In this work, the 4th China Physiological Signal Challenge (CPSC) 2021 dataset [9] was employed, which provides a two-stage training set. The first stage contains 730 ECG recordings collected from 12 AF subjects and 42 non-AF subjects, whereas the second stage includes 706 recordings obtained from 37 AF and 14 non-AF subjects. For the purpose of this study, a subset was selected, consisting of 20 recordings for model training (7 AF and 13 non-AF) and 10 recordings for evaluation (5 AF and 5 non-AF). Fig. 1 illustrate a normal ECG record and an AF record, respectively. The AF record demonstrates abnormality through irregular RR intervals and disrupted waveform morphology, distinguishing it from the normal rhythm.

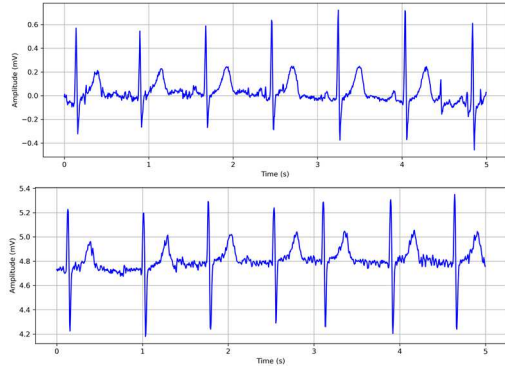


Fig. 1. The example of Normal (top) and AF(bottom) ECG signal.

B. Methodology

The overall architecture of the proposed AF detection framework is illustrated in Fig. 2. It is organized into three stages, beginning with signal preprocessing and segmentation, followed by the extraction of meaningful features, and finally the classification of AF episodes using machine learning techniques.

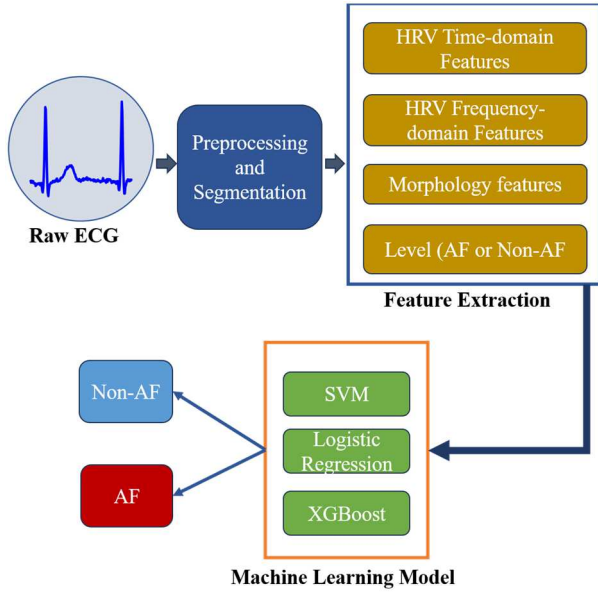


Fig. 2. Block diagram of the overall architecture of the proposed AF detection framework.

C. Preprocessing and Segmentation:

The ECG recording was preprocessed and divided into overlapping of 20 seconds window. Segmenting the signals into shorter intervals allows for localized examination of rhythm variations and improves the ability of the model to capture AF episodes that may occur intermittently. The choice of a 20-second window provides a balance between retaining sufficient temporal information for feature extraction and reducing computational complexity during training and evaluation. Each segmented window was subsequently processed for feature extraction and labeling, forming the basis for the classification task.

D. Feature Extraction

In this study, three categories of features were extracted—HRV time-domain, HRV frequency-domain, and ECG morphology-based parameters—using the NeuroKit2 toolbox [10]. The detailed list of the extracted features is provided in TABLE I.

TABLE I. FEATURE EXTRACTED FROM EACH ECG SEGMENTS

Category	Features	Description
HRV (Time domain)	HRV_MeanNN	Mean of normal-to-normal (NN) intervals
	HRV_SDNN	Standard deviation of NN intervals
	HRV_SDANN1, SDANN2, SDANN5	SD of NN intervals over 1, 2, 5 min windows
	HRV_SDNNI1, SDNNI2, SDNNI5	Mean of SD of NN intervals within 1, 2, 5 min windows
	HRV_RMSSD	Root mean square of successive differences
	HRV_SSDSD	Standard deviation of successive differences
	HRV_CVNN	Coefficient of variation of NN intervals
	HRV_CVSD	Coefficient of variation of successive differences
	HRV_MedianNN, MadNN, MCVNN, IQRNN	Median, Median Absolute Deviation, Median Coefficient of Variation and Interquartile Range measures of NN intervals
	HRV_SDRMSSD	Ratio of SDNN to RMSSD
	HRV_Prc20NN, Prc80NN	20th and 80th percentiles of NN intervals
	HRV_pNN50	% of successive NN intervals > 50 ms
	HRV_pNN20	% of successive NN intervals > 20 ms
	HRV_MinNN, MaxNN	Shortest and longest NN intervals
HRV (Frequency domain)	HRV_HTI	HRV Triangular Index
	HRV_TINN	Baseline width of NN interval histogram
	ULF, VLF, LF, HF, VHF	Ultra-low, very low, low, high, and very high frequency power
	TP	Total power
Morphology	LF/HF	Ratio of LF to HF
	LFn, HFn	Normalized LF and HF
	LnHF	Log-transformed HF power
	Morph_Label	Class label for morphology epoch
	Morph_Event_Onset	Onset timing of detected event
	Morph_ECG_Rate_Baseline	Baseline heart rate
	Morph_ECG_Rate_Max	Maximum heart rate in the segment
	Morph_ECG_Rate_Min	Minimum heart rate in the segment
	Morph_ECG_Rate_Mean	Mean heart rate in the segment

E. Machine Learning Model

To automatically classify AF from normal segment, there different machine learning algorithms can be employed, including SVM, LR, and XGBoost.

Support Vector Machine (SVM): SVM is a prominent machine learning technique designed to find the hyperplane that separates data points in an n-dimensional space. It can tackle non-linear classification challenges by using different kernel functions. The core principle of SVM is to establish a decision boundary that differentiates data points.

The vector w denotes the normal to the separating hyperplane, while u represents the input feature vectors. The parameter b acts as the bias term, defining the offset of the hyperplane. The separation constraints for the positive and negative classes are expressed in (1) and (2), respectively:

$$w \cdot u + b > 0 \quad (1)$$

$$w \cdot u + b < 0 \quad (2)$$

Logistic Regression: Commonly referred to as logit regression, this statistical approach is employed to estimate the likelihood of an event based on historical data. Logistic regression is versatile, functioning effectively with both binary and multi-class data to assess whether an event takes place or not.

XGBoost: XGBoost is a well-regarded classifier that has been extensively applied for the detection of atrial fibrillation (AF), showcasing outstanding performance on extensive datasets. Its high degree of flexibility allows it to address numerous problems in regression, classification, and ranking. For utilizing XGBoost, we utilized the scikit-learn package to train the classifier.

III. RESULT AND DISCUSSION

Three classifiers—SVM, Logistic Regression, and XGBoost—were evaluated in the proposed model. The SVM achieved 96.1% accuracy for Non-AF segments and 95% for AF segments, with an overall accuracy of 95.4%. Logistic Regression yielded 94.17% and 93.57% accuracy for Non-AF and AF segments, respectively, with an overall accuracy of 93.8%, while XGBoost achieved 96.1% for Non-AF and 94.28% for AF segments, with an overall accuracy of 95%. These results demonstrate the robustness of the proposed approach across classifiers. The corresponding confusion matrices are shown in Fig. 3.

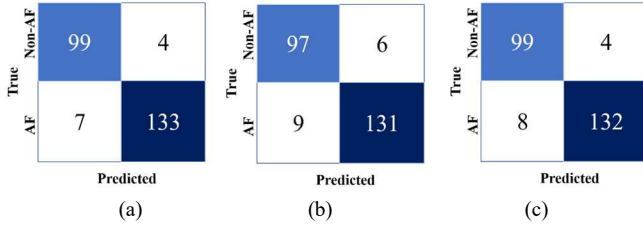


Fig. 3. The confusion Matrix for AF classification using ECG signal for (a) SVM (b) Logistic Regression (c) XGBoost Classifier

The Receiver Operating Characteristic (ROC) curves demonstrate that the proposed classifiers achieve strong performance in distinguishing AF from Non-AF cases. All classifiers exhibit high discriminative capability, with Area Under the Curve (AUC) values approaching 0.99, underscoring the robustness of the proposed framework in reliably separating AF and Non-AF episodes, as shown in Fig. 4(a). Among them, the SVM achieved the best performance, reflecting its superior generalization capability. Similarly, the Precision-Recall (PR) curves confirm that all classifiers perform well in differentiating AF from Non-AF cases. Each model achieved high precision alongside substantial recall, with SVM again showing the most balanced trade-off, indicating effective detection of AF instances while minimizing false positives. These findings further validate the reliability of the proposed framework, as illustrated in Fig. 4(b).

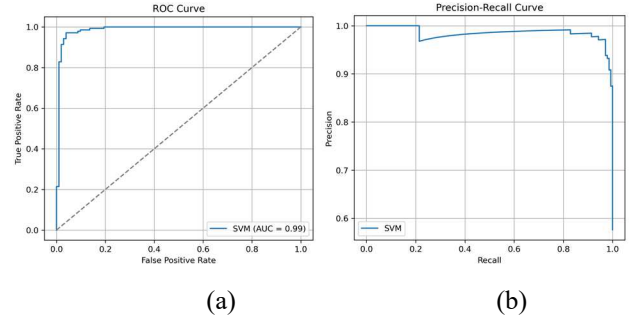


Fig. 4. (a) The ROC Curve and (b) the Precision-Recall Curve of the proposed AF detection model

The decision boundary of the proposed classifier demonstrates a clear separation between AF and Non-AF classes in the feature space. AF samples form a distinct cluster apart from Non-AF samples, while most misclassified points lie near the separating margin, suggesting they correspond to borderline or noisy patterns rather than systematic errors. This visualization confirms the effectiveness of the extracted features and the classifier in discriminating AF from Non-AF, as illustrated in Fig. 5.

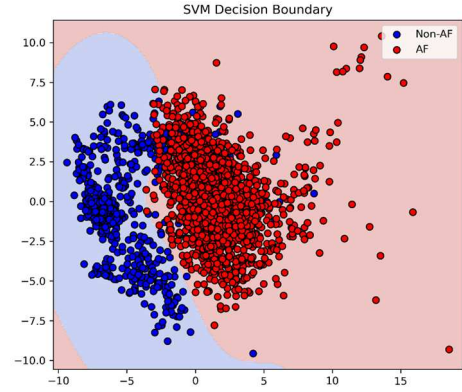


Fig. 5. SVM Decision Boundary of proposed AF detection model

The relative importance of the extracted features was evaluated using the XGBoost model, revealing that HRV-related features dominate the classification. HRV_HRV_MCVNN contributed the highest importance (0.37), followed by HRV_HRV_MadNN (0.23) and HRV_HRV_IQRNN (0.06), while morphological ECG features such as ECG_Rate_Max and ECG_Rate_Min_Time had comparatively lower contributions. These results highlight that temporal variability in RR intervals is the most discriminative factor for distinguishing AF from Non-AF, with morphological descriptors providing complementary information, as shown in Fig. 6.

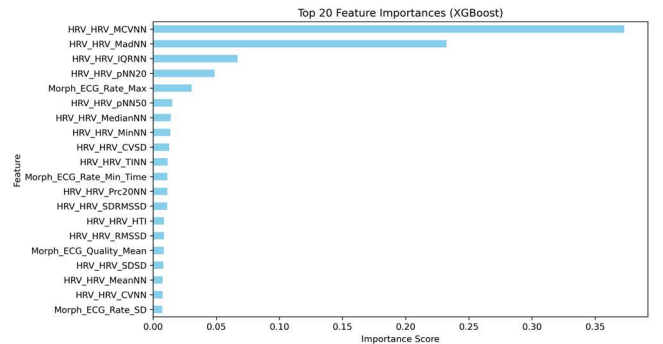


Fig. 6. Top 20 Feature Importance using XGBoost. of proposed AF detection model

TABLE II. CLASSIFICATION REPORT FOR DIFFERENT CLASSIFIER

Model	Precision	Recall	F1-Score	Accuracy
SVM	0.953	0.954	0.956	0.954
Logistic Regression	0.943	0.94	0.941	0.938
XGBoost	0.953	0.948	0.951	0.95

The classification outcomes of the proposed models in distinguishing AF from Non-AF segments show that the SVM classifier achieved the highest overall accuracy of 95.4%, with precision, recall, and F1-score of 0.953, 0.954, and 0.956, respectively, indicating balanced performance across both classes. Logistic Regression yielded slightly lower results, with an accuracy of 93.8%, and precision, recall, and F1-score of 0.943, 0.940, and 0.941. The XGBoost model achieved an overall accuracy of 95.0%, accompanied by precision, recall, and F1-score values of 0.953, 0.948, and 0.951, showing strong but slightly lower recall for the AF class compared to SVM. These results confirm the robustness of the proposed features across different classifiers and highlight the superior generalization capability of SVM for AF detection, as detailed in TABLE II.

Overall, the proposed framework demonstrates effective AF detection using ECG-derived features. Feature importance analysis indicates that HRV-based indices, particularly variability measures such as MCVNN and MadNN, are the most discriminative, while morphological descriptors provide complementary information. ROC and PR analyses further confirm strong discriminative capability across classifiers, with AUC values approaching 0.99. Various machine learning and deep learning models have been explored in recent studies for AF detection, achieving accuracies in the range of 87–95%. The proposed framework was found to achieve an overall accuracy of 95%, demonstrating competitive or superior performance, as summarized in TABLE III. Future work will focus on validation with larger datasets to improve generalizability.

TABLE III. THE PERFORMANCE COMPARISON OF THE PROPOSED MODEL WITH THE EXISTING MODEL IN THE LITERATURE.

Reference	Model	Prec	Rec	F1	Acc
Liaqat et al. [11]	CNN + LSTM	0.86	0.85	0.86	0.87
Warrick et al. [12]	CNN + LSTM	0.77	0.76	0.76	0.77
Akhoondkazemi et al. [13]	DL+GAN	0.94	0.95	0.95	0.95
Wang et al. [14]	CNN+LSTM	0.93	0.94	0.94	0.94
Rubin et al. [15]	CNN	0.80	0.79	0.80	0.80
Proposed	SVM	0.95	0.95	0.96	0.95

IV. CONCLUSION

This paper presented a machine learning-based approach for detecting atrial fibrillation from ECG recordings. A wide set of HRV and morphological features were extracted from ECG signal, the proposed framework achieved robust performance using SVM classifier the highest accuracy of 95.4%. The feature importance analysis showed that irregularities in RR intervals are the most reliable indicators of AF, while ECG morphology provides additional support. These outcomes confirm that a feature-driven approach can

offer both accuracy and interpretability for AF detection. As a next step, we aim to expand this work by using the full CPSC 2021 dataset to train and evaluate the models on a larger and more diverse population. We also plan to explore more advanced classifiers, such as ensemble methods and deep learning architectures, to further boost detection performance. Finally, we envision adapting this framework for wearable and real-time monitoring systems, making AF detection more accessible in clinical and home settings.

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